

Hidden States

$Z_{in} \in \{C, H, N\}$
 i = game index
 n = shot num or
 Z_{in} = hidden state during shot n
 Y_{in} = observed shot outcome, made or missed $\in \{0, 1\}$

Transition Matrix

Stores probabilities for going from state (C, H, N) to state (C, H, N) .
Row = current state
Column = next state
Looks like $P = \begin{pmatrix} p(C|C) & p(C|N) & p(C|H) \\ p(N|C) & p(N|N) & p(N|H) \\ p(H|C) & p(H|N) & p(H|H) \end{pmatrix}$
and each row should sum to 1.

Transition dist.

$Z_{i,n+1} | Z_{in} \sim \text{Categorical}(p_i(C|C), p_i(C|N), p_i(C|H))$, $j \in \{C, H, N\}$

Softmax regression

Used for state transition probabilities:
 $j \in \{C, H, N\}$
 $\eta_i(j|C) = \beta(j|C) + b_i(j|C)$
 $\eta_i(j|H) = \beta(j|H) + b_i(j|H)$
 $\eta_i(j|N) = \beta(j|N) + b_i(j|N)$
 $P_i(j|C) = \frac{e^{\eta_i(j|C)}}{1 + e^{\eta_i(C|C)} + e^{\eta_i(C|H)} + e^{\eta_i(C|N)}}$
 $P_i(j|H) = \frac{e^{\eta_i(j|H)}}{1 + e^{\eta_i(H|C)} + e^{\eta_i(H|H)} + e^{\eta_i(H|N)}}$
 $P_i(j|N) = \frac{1}{1 + e^{\eta_i(N|C)} + e^{\eta_i(N|H)} + e^{\eta_i(N|N)}}$

Shot prob. regression

$\logit(Y_i(C)) = \beta(C) + b_i(C)$
 X_i = design matrix of observed features in game i .
Ex. $X_i = \begin{bmatrix} 1 & \text{distance}_1 & \text{time}_1 \\ 1 & \text{distance}_2 & \text{time}_2 \\ 1 & \text{distance}_3 & \text{time}_3 \end{bmatrix}$
where each row is a shot
 $\alpha(C)$ = fixed-effect coefficients

Initial Distribution

$S_i = (S_i(C), S_i(N), S_i(H))$
Starting with $S_i \sim \text{Dirichlet}(1, 1, 1)$
so all states equally likely at start of game i .
If a player is more likely to start cold adjust to $\text{Dirichlet}(5, 1, 1)$ so cold state is higher prob.

Stationary Distribution

Stationary distribution will be the limit distribution.
 $\Delta i = (\Delta i^C, \Delta i^H, \Delta i^N)$, which turns out to be
 $\Delta i^C = \frac{P_i^{(H|C)}}{P_i^{(C|H)} + P_i^{(H|H)} + P_i^{(H|N)}}$
 $\Delta i^H = \frac{P_i^{(C|H)}}{P_i^{(C|H)} + P_i^{(H|H)} + P_i^{(N|H)}}$
 $\Delta i^N = \frac{P_i^{(N|H)}}{P_i^{(C|H)} + P_i^{(H|H)} + P_i^{(N|H)}}$

3.2 t-step transition probabilities and Stationary distribution

t -step transition probabilities of the chain for game i :
 $P_i^{(H|H)}(t) = P(Z_{i,n+1} = H | Z_{in} = H, \theta, \psi)$
 $P_i^{(C|H)}(t) = P(Z_{i,n+1} = C | Z_{in} = H, \theta, \psi)$
 $P_i^{(N|H)}(t) = P(Z_{i,n+1} = N | Z_{in} = H, \theta, \psi)$
These are basically saying: "What state will we be in t shots from now?" Ex: we have $P_i^{(C|H)}(t)$ probability to go from state C to state H in t shots.
Matrix $P_i^{(H|H)}$ that collects these probabilities can be calculated from the transition probability matrix P_i shown below:
 $P_i^{(H|H)} = \frac{1}{P_i^{(H|H)} + P_i^{(H|C)} + P_i^{(H|N)}} \begin{bmatrix} P_i^{(H|H)} & P_i^{(H|C)} & P_i^{(H|N)} \\ P_i^{(C|H)} & P_i^{(C|C)} & P_i^{(C|N)} \\ P_i^{(N|H)} & P_i^{(N|C)} & P_i^{(N|N)} \end{bmatrix}$
definitions
HMC = Hidden Markov Chain. Finite state space: we only have 2 states (in the paper not listed) that a shooter can be in (which is a finite number).
Irreducible: It's possible to get from one state to any other state eventually. We can't "reduce" the number of states we can visit if we get to a certain state.
State = periodic if chain can only return to the state at certain predictable intervals (ex only on every 3rd shot). Aperiodic = No fixed cycle.

3.1 Transition Probabilities

Posterior Distribution = $\pi(\theta, \psi | D)$: Joint Prob of distribution of all the unknowns in the model given your data (D).
 D = data, includes the observations y , ψ and a few covariates.
 ψ = random effects
 θ = parameters and hyperparameters
In 2.0, $\beta^{(H|H)}$ and $\beta^{(H|C)}$ = Regression coefficient vectors associated with transitions from C to H and H to C respectively.
 ψ = random effects associated with transition probabilities (β). Assumed to follow a conditional multivariate normal distribution as $(\beta, \psi) \sim N(\mu, \Sigma)$, where Σ is a variance-covariance matrix.
Note: β and Σ are parameters and hyperparameters included in the generic θ vector. Same applies to the random effects ψ 's as elements of ψ .

3.3 Occupancy Times

Refers to the # of times a state (K) is visited in a given # of transitions given the starting state was j .
Expected value of occupancy time / expected # of shots in state (K time):
 $V_i^{(H|H)}(t) = E(V_i^{(H|H)}(t) | \theta, \psi)$
our counter (V).
 $V_i^{(H|H)}(t)$: The number of times state K is visited given the starting state in the chain is j over t shots.
Note: V is an occupancy counter, not an arrival counter. V counts the # of shots that are in a state (kind of I think), not the # of times a team plays more into a state.

These conditional expectations are expressed in terms of the transition probabilities as follows:
 $V_i^{(H|H)}(t) = \frac{t+1}{P_i^{(H|H)} + P_i^{(H|C)} + P_i^{(H|N)}} \begin{bmatrix} P_i^{(H|H)} & P_i^{(H|C)} \\ P_i^{(C|H)} & P_i^{(C|C)} \end{bmatrix} + \frac{1 - (P_i^{(H|H)} + P_i^{(H|C)} + P_i^{(H|N)})^{t+1}}{(P_i^{(H|H)} + P_i^{(H|C)} + P_i^{(H|N)})^2} \begin{bmatrix} P_i^{(H|H)} & P_i^{(H|C)} \\ P_i^{(C|H)} & P_i^{(C|C)} \end{bmatrix}$

Completing the posterior distribution of each occupancy time gives us information on each game. In a general way, we can integrate over the random effects associated with the individual games and obtain posterior information of the # of visits among a given # transitions (shots) in a generic game.

3.4 Sojourn Times

Number of shots required for team/player to transition out of state j (for the first time in this paper).
 $j = \text{team}, j \in \{C, H\}$, $i = \text{game}, i \in \{1, 2, \dots, N\}$
 $P_i^{(H|H)}(t) = P(Z_{i,n+1} = H | Z_{in} = H, \theta, \psi) = \sum_{j=C, N} P_i^{(H|j)}(t) = \sum_{j=C, N} P_i^{(H|j)}(t) = \sum_{j=C, N} P_i^{(H|j)}(t)$
The arbitrary value (the input) we set equal to the theoretical # of shots we want to test. We plug this value into the above equation to find the probability of a streak lasting exactly t^* shots in state j (in pretty sure).
Conditional distribution of the sojourn time in state C (H) of game i will be a geometric distribution of parameter $P_i^{(C|H)}$ ($P_i^{(H|H)}$). (?)
Posterior distribution of $\tau_i^{(C)}$ and $\tau_i^{(H)}$:
 $\pi(\tau_i^{(C)} = n | \theta, \psi | D)$
 $\pi(\tau_i^{(H)} = n | \theta, \psi | D)$

Marginal posterior distribution can be found by integrating out the random effects.

3.5 Probabilities of Making a Basket

How this applies to the Hot Hand situation
This linear algebra property is the formal proof for why the "memory" of a streak eventually vanishes. The vector Δi represents the long-term equilibrium of the Miami Heat (which we established is roughly a 39% chance of being Hot and a 61% chance of being Cold).
Saying that Δi is the left eigenvector with an eigenvalue of 1 is just the mathematician's way of saying:
"Once the team's probability of being hot vs. cold reaches this exact baseline balance (Δi), taking another shot (Πi) will not disrupt that balance. The team has reached statistical equilibrium."
It guarantees that the "Limit Distribution" we talked about earlier actually exists and is completely stable.

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